Afzal Hussain

afzalhussain@iut-dhaka.edu — afzalhussain23.github.io/research

EDUCATION

Islamic University of Technology IUT

2018 **B.S.**, Mechanical Engineering; CGPA: **3.70**/4.00
Gazipur, Bangladesh
Thesis: *Multi-Objective Optimization of Turning Operation of Stainless Steel Using a Hybrid Whale Optimization Algorithm.*
Advisor: Prof. Mohammad Ahsan Habib

RESEARCH EXPERIENCE

My research interests are scientific machine learning for computational mechanics, data-driven surrogate modeling for physics-based engineering systems, and design optimization, with growing interest in transformer and diffusion-based methods for topology optimization and design exploration.

Self-Directed Research
Computational Fluid Dynamics

Machine Learning for Transient CFD Simulation
Generated an OpenFOAM dataset of approximately 1,500 transient Mach-3 forward-facing-step simulations with parameterized geometries. Built an end-to-end simulation-to-ML pipeline for preprocessing CFD fields, training a ConvLSTM-ResNet surrogate model, and visualizing field predictions. Evaluated the model on unseen geometries to predict velocity, pressure, and temperature fields for compressible-flow simulations. GitHub Link

Islamic University of Technology
Gazipur, Bangladesh

Multi-Objective Engineering Optimization
Developed and evaluated a hybrid Whale Optimization Algorithm for multi-objective optimization of stainless-steel turning processes. Applied metaheuristic optimization to engineering design/process parameters, with emphasis on balancing competing manufacturing objectives. This work was peer-reviewed and published in the *Journal of Manufacturing and Materials Processing.*

PUBLICATIONS

- Multi-Objective Optimization of Turning Operation of Stainless Steel Using a Hybrid Whale Optimization Algorithm. Mahamudul Hasan Tanvir, **Afzal Hussain**, M. M. Towfiqur Rahman, Sakib Ishraq, Khandoker Zishan, SK Tashowar Tanzim Rahul, and Mohammad Ahsan Habib. *Journal of Manufacturing and Materials Processing*, 4(3), 64, 2020. https://doi.org/10.3390/jmmp4030064

- Effectiveness of Hard Clustering Algorithms for Securing Cyber Space. Sakib Mahtab Khandaker, **Afzal Hussain**, and Mohiuddin Ahmed. In *Smart Grid and Internet of Things*, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, pp. 113–120. Springer International Publishing, 2019. https://doi.org/10.1007/978-3-030-05928-6_11

INDUSTRY EXPERIENCE

Pathao
Dhaka, Bangladesh
03/2023 – 11/2025
Senior Software Engineer
Led modernization of large-scale distributed systems serving over 10 million users, with emphasis on scalable computation, reliable workflows, observability, and data-intensive software. Built durable asynchronous systems for failure-prone processes, improving experience in testing, debugging, production reliability, and maintainable computational infrastructure.

Hogarth
Dhaka, Bangladesh
11/2021 – 02/2023
Software Engineer
Developed enterprise software systems for global clients across multiple deployment environments, with emphasis on modular implementation, testing, debugging, and maintainability. Led investigations of production issues using structured root-cause analysis and careful validation.

Enosis Solutions
Dhaka, Bangladesh
12/2018 – 10/2021
Software Engineer
Built database-backed modules for point-of-sale, order fulfillment, inventory synchronization, reporting, and real-time communication systems. Gained experience in data modeling, integration workflows, SQL-backed applications, and reliable end-to-end software delivery.

AWARDS AND HONORS

- **National finalist** in the Bangladesh Physics Olympiad (BdPhO) and Bangladesh Olympiad on Astronomy and Astrophysics (BDOAA), 2013.

- Placed **18th** nationally in the Bangladesh Olympiad in Informatics (BdOI), 2014.

- Ranked **2nd** in the IUT intra-university competitive programming team selection; member of the **Best Freshman Team** from IUT at the ICPC Dhaka Regional, 2015.

TECHNICAL SKILLS

- Programming and scripting: Python, C/C++, MATLAB, SQL, Bash/Linux.
- Machine learning and scientific computing: PyTorch, TensorFlow/Keras, NumPy, SciPy, pandas, scikit-learn, Matplotlib, Jupyter Notebook.
- Engineering simulation and optimization: OpenFOAM, computational fluid dynamics (CFD), computer-aided design (CAD), multi-objective optimization, data-driven surrogate modeling.
- Computational research workflows: simulation data generation, preprocessing, model training, visualization, reproducible pipelines, Git, Docker.
- Software systems: distributed systems, event-driven architecture, PostgreSQL, Kubernetes, AWS, observability, production reliability.
- Language models: Transformer architectures, attention mechanisms, tokenization, language-model implementation and evaluation, LLM-assisted development.

LEADERSHIP AND SERVICE

- Coached university students in algorithms and data structures for competitive programming contests.
- Executive Member, IUT Computer Society; helped organize technical events and student programming initiatives.
- Mentored software engineers through code reviews, onboarding, architecture discussions, and distributed-systems design reviews.

SELECTED COURSEWORK

- Language Modeling from Scratch (CS336), Stanford University, in progress.
- Deep Learning Specialization, Coursera.
- Machine Learning, Coursera.
- A Hands-on Introduction to Engineering Simulations, edX.
- Using Python for Research, edX.